

TIMER BEHAVIOR SCRIPT INTERFACE

View online: <https://www.construct.net/en/make-games/manuals/construct-3/scripting/scripting-reference/behavior-interfaces/timer>

The `ITimerBehaviorInstance` interface derives from `IBehaviorInstance` to add APIs specific to the **Timer behavior**.

Example

Below is a sample code snippet demonstrating setting two timers on `inst` (assumed to be an instance with the Timer behavior). One is a regular timer and the other is a one-off timer.

Note that many browsers combine identical log messages in the console, so repeat messages may just increment a number by the message rather than showing it again.

```
// Handle "timer" event and log to the console that the timer ran.
inst.behaviors.Timer.addEventListener("timer", e =>
{
    console.log(`Timer "${e.tag}" ran!`);
});

// Set a regular timer running every 1 second
inst.behaviors.Timer.startTimer(1, "myRegularTimer", "regular");

// Set a one-off timer to run after 2.5 seconds
inst.behaviors.Timer.startTimer(2.5, "myOneOffTimer", "once");
```

Timer behavior events

See **behavior instance event** for standard behavior instance event object properties.

"timer"

Fired when a timer period has elapsed. The event object has a `tag` property which is a string of the tag for the timer that has elapsed.

Timer behavior APIs

startTimer(duration, tag, type = "once")

Set a new timer, or if a timer with the same `tag` exists, re-start it with new options.

`duration` is the time in seconds until the `"timer"` event fires. If `type` is `"once"`, then the timer event will fire once and not again until `startTimer` is called again. If `type` is set to `"regular"`, then the timer event will keep firing every `duration` seconds. The `tag` is a string that allows identifying different timers.

setTimerPaused(tag, isPaused)

Set a currently running timer either paused or resumed. When a timer is paused, it will stop firing timer events. When resumed, it will continue firing timer events, resuming from the time that it was paused at. In other words if a timer is set for 1 second, it is paused after 0.5 seconds, and then after some time it is resumed again, the next timer event will fire 0.5 seconds after resuming.

setAllTimersPaused(isPaused)

This does the same thing as the `setTimerPaused()` method, but affecting all existing timers rather than only one with a given tag.

stopTimer(tag)

Stop a timer with a specific tag. The timer event will no longer fire for the stopped timer after this call.

stopAllTimers()

Stop all currently running timers regardless of their tags. The timer event will no longer fire for any timer after this method unless a new timer is started.

isTimerRunning(tag)

Returns a boolean indicating if a timer with the given tag has been started. Once stopped, the timer no longer counts as running. Paused timers also count as running - use `isTimerPaused()` to identify these timers separately.

isTimerPaused(tag)

Returns a boolean indicating if a timer with the given tag has been started and then subsequently paused with `setTimerPaused()`.

getCurrentTime(tag)

Returns the time in seconds since the `"timer"` event last fired, for a timer with a specific tag.

getTime(tag)

Returns the time in seconds since a timer with a specific tag was started. This is only useful with regular timers, since it will always equal `getCurrentTime()` for one-off timers (after which they fire and the timer no longer exists, so these expressions return 0).

getDuration(tag)

Returns the duration in seconds for a timer with a specific tag.

getNormalizedProgress(tag)

Returns the current progress for a timer with a specific tag represented as a number between 0 and 1, regardless of the duration of the timer. For example a normalized progress of 0.5 means half way through the duration.

hasFinished(tag)

Returns a boolean that is true for the tick that the `"timer"` event fires in. This is the way the *On timer* condition of the Timer behavior checks for finished timers, and this method allows script callers to make the same check, such as by polling the timer in the `"tick"` event.